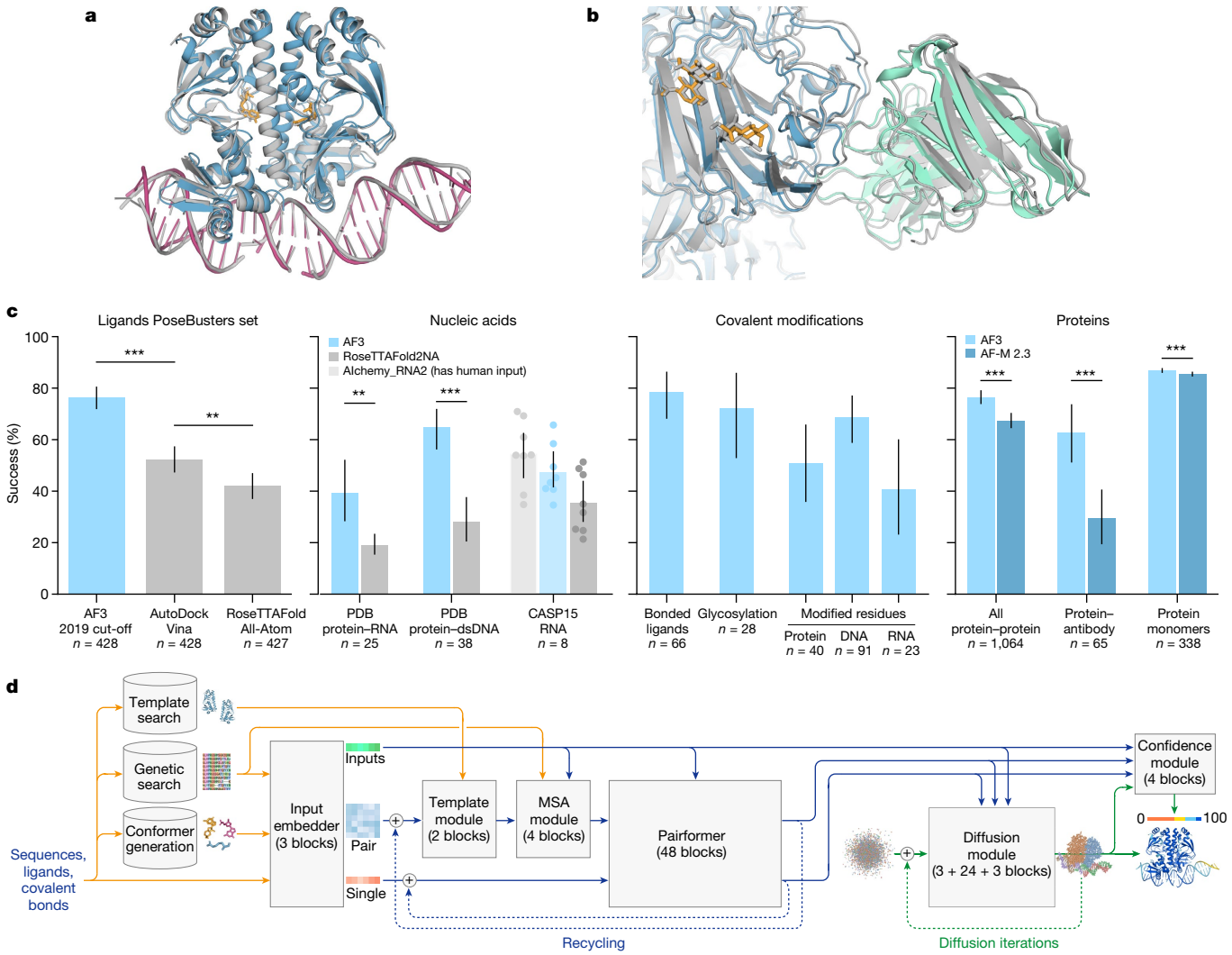


Fig. 1 | AF3 accurately predicts structures across biomolecular complexes. **a**,**b**, Example structures predicted using AF3. **a**, Bacterial CRP/FNR family transcriptional regulator protein bound to DNA and cGMP (PDB 7PZB; full-complex LDDT[47], 82.8; global distance test (GDT)[48], 90.1). **b**, Human coronavirus OC43 spike protein, 4,665 residues, heavily glycosylated and bound by neutralizing antibodies (PDB 7PNM; full-complex LDDT, 83.0; GDT, 83.1). **c**, AF3 performance on PoseBusters (v.1, August 2023 release), our recent PDB evaluation set and CASP15 RNA. Metrics are as follows: percentage of pocket-aligned ligand r.m.s.d. < 2 Å for ligands and covalent modifications; interface LDDT for protein–nucleic acid complexes; LDDT for nucleic acid and protein monomers; and percentage DockQ > 0.23 for protein–protein and protein–antibody interfaces. All scores are reported from the top confidence-ranked sample out of five model seeds (each with five diffusion samples), except for protein–antibody scores, which were ranked across 1,000 model seeds for both models (each AF3 seed with five diffusion samples). Sampling and ranking details are provided in the Methods. For ligands, *n* indicates the number of targets; for nucleic acids, *n* indicates the number of structures; for modifications, *n* indicates the number of clusters; and for proteins, *n* indicates the number of clusters. The bar height indicates the mean; error bars indicate exact binomial distribution 95% confidence intervals for PoseBusters and by 10,000 bootstrap resamples for all others. Significance levels were calculated using two-sided Fisher's exact tests for PoseBusters and using two-sided Wilcoxon signed-rank tests for all others; $^{***}P < 0.001$, $^{**}P < 0.01$. Exact *P* values (from left to right) are as follows: 2.27×10^{-13}, 2.57×10^{-3}, 2.78×10^{-3}, 7.28×10^{-12}, 1.81×10^{-18}, 6.54×10^{-5} and 1.74×10^{-34}. AF-M 2.3, AlphaFold-Multimer v.2.3; dsDNA, double-stranded DNA. **d**, AF3 architecture for inference. The rectangles represent processing modules and the arrows show the data flow. Yellow, input data; blue, abstract network activations; green, output data. The coloured balls represent physical atom coordinates.

Network architecture and training

The overall structure of AF3 (Fig. 1d and Supplementary Methods 3) echoes that of AF2, with a large trunk evolving a pairwise representation of the chemical complex followed by a structure module that uses the pairwise representation to generate explicit atomic positions, but there are large differences in each major component. These modifications were driven both by the need to accommodate a wide range of chemical entities without excessive special casing and by observations of AF2 performance with different modifications. Within the trunk, MSA processing is substantially de-emphasized, with a much smaller and simpler MSA embedding block (Supplementary Methods 3.3). Compared with the original evoformer from AF2, the number of blocks is reduced to four, the processing of the MSA representation uses an inexpensive pair-weighted averaging and only the pair representation is used for later processing steps. The 'pairformer' (Fig. 2a and Supplementary Methods 3.6) replaces the evoformer of AF2 as the dominant processing block. It operates only on the pair representation and the single representation; the MSA representation is not retained and all information passes through the pair representation. The pair processing and the number of blocks (48) is largely unchanged from AF2. The resulting pair and single representation together with the input representation are passed to the new diffusion module (Fig. 2b) that replaces the structure module of AF2.

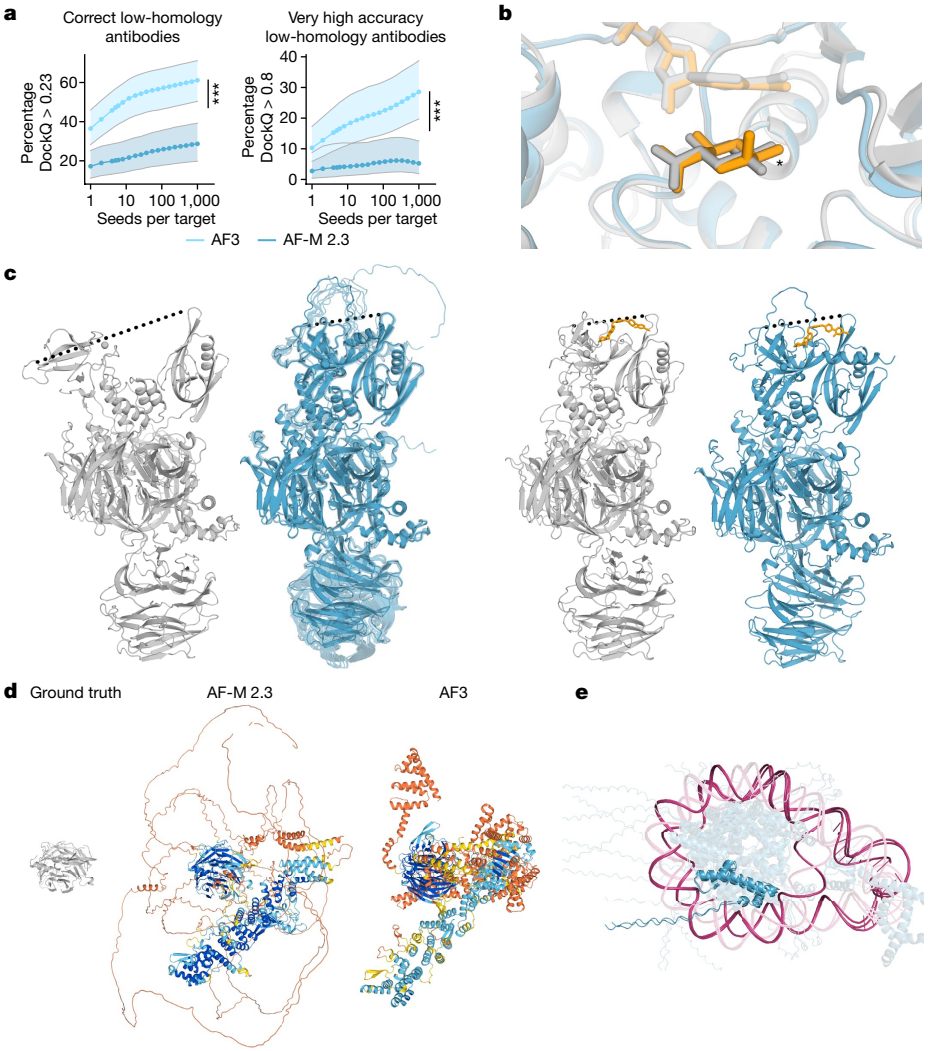


Fig. 5 | Model limitations. a, Antibody prediction quality increases with the number of model seeds. The quality of top-ranked, low-homology antibody–antigen interface predictions as a function of the number of seeds. Each datapoint shows the mean over 1,000 random samples (with replacement) of seeds to rank over, out of 1,200 seeds. Confidence intervals are 95% bootstraps over 10,000 resamples of cluster scores at each datapoint. Samples per interface are ranked by protein–protein ipTM. Significance tests were performed using by two-sided Wilcoxon signed-rank tests. $n = 65$ clusters. Exact P values were as follows: 2.0×10^{-5} (percentage correct) and $P = 0.009$ (percentage very high accuracy). **b**, Prediction (coloured) and ground-truth (grey) structures of *Thermotoga maritima* α-glucuronidase and beta-D-glucuronic acid—a target from the PoseBusters set (PDB: 7CTM). AF3 predicts alpha-D-glucuronic acid; the differing chiral centre is indicated by an asterisk. The prediction shown is top-ranked by ligand–protein ipTM and with a chirality and clash penalty. **c**, Conformation coverage is limited. Ground-truth structures (grey) of cereblon in open (apo, PDB: 8CVP; left) and closed (holo mezigdomide-bound, PDB: 8D7U; right) conformations. Predictions (blue) of both apo (with 10 overlaid samples) and holo structures are in the closed conformation. The dashed lines indicate the distance between the N-terminal Lon protease-like and C-terminal thalidomide-binding domain. **d**, A nuclear pore complex with 1,854 unresolved residues (PDB: 7F60). The ground truth (left) and predictions from AlphaFold-Multimer v.2.3 (middle) and AF3 (right) are shown. **e**, Prediction of a trinucleosome with overlapping DNA (pink) and protein (blue) chains (PDB: 7PEU); highlighted are overlapping protein chains B and J and self-overlapping DNA chain AA. Unless otherwise stated, predictions are top-ranked by our global complex ranking metric with chiral mismatch and steric clash penalties (Supplementary Methods 5.9.1).

strongly is antibody–antigen complexes, similar to other recent work[46]. Figure 5a shows that, for AF3, top-ranked predictions keep improving with more model seeds, even at as many as 1,000 (Wilcoxon signed-rank test between 5 and 1,000 seeds, $P = 2.0 \times 10^{-5}$ for percentage correct and $P = 0.009$ for percentage very high accuracy; ranking by protein–protein interface ipTM). This large improvement with many seeds is not observed in general for other classes of molecules (Extended Data Fig. 7b). Using only one diffusion sample per model seed for the AF3 predictions rather than five (not illustrated) does not change the results significantly, indicating that running more model seeds is necessary for antibody score improvements, rather than just more diffusion samples.

Discussion

The core challenge of molecular biology is to understand and ultimately regulate the complex atomic interactions of biological systems. The AF3 model takes a large step in this direction, demonstrating that it is possible to accurately predict the structure of a wide range of biomolecular systems in a unified framework. Although there are still substantial challenges to achieve highly accurate predictions across all interaction types, we demonstrate that it is possible to build a deep-learning system that shows strong coverage and generalization for all of these interactions. We also demonstrate that the lack of cross-entity evolutionary information is not a substantial blocker to progress in predicting